

EC1207 — Principles of the Micro Economy

Winter 2024 · Paper 1

MARKING SCHEME

Module	EC1207 — Principles of the Micro Economy
Exam Session	Winter 2024 · Paper 1
Duration	1.5 Hours
Internal Examiners	Dr Tracy Bradfield
Head of School	Prof. Justin Doran
External Examiner	Dr Kevin Mulligan

Section	Format	Weight
A	Short Questions — attempt 2 out of 3	40%
B	Long Questions — attempt 1 out of 2	60%

SECTION A — SHORT QUESTIONS (Attempt 2 out of 3 · 40% of total marks)

Each full question is worth 20%. Each sub-part is worth 10%. Target approximately 1.5 pages per 10% sub-part. Include 2 academic references per page, cited as (Author, Year, Publication).

A1 (i)

10 percent

Using the supply and demand model, illustrate the effects on market equilibrium of: (a) an increase in the number of suppliers, and (b) an increase in the price of substitutes.

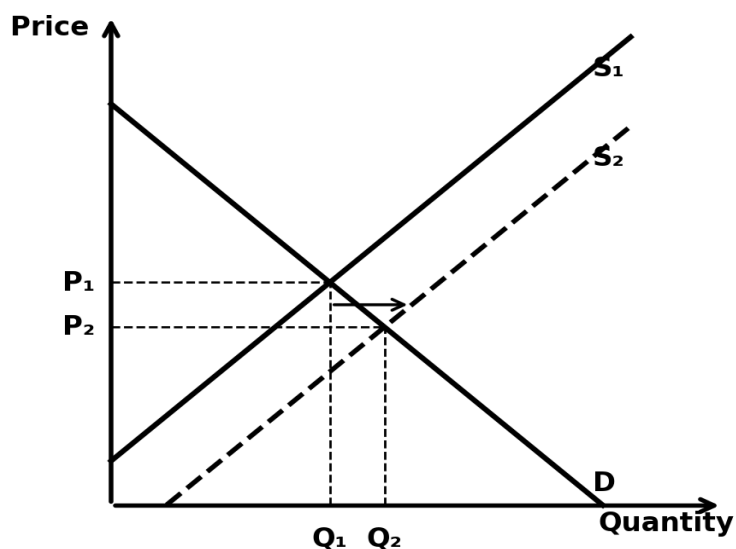
(a) More Suppliers → Supply Shifts Right

Figure 1(a): More suppliers shift S rightward — equilibrium price falls ($P_1 \rightarrow P_2$), equilibrium quantity rises ($Q_1 \rightarrow Q_2$).

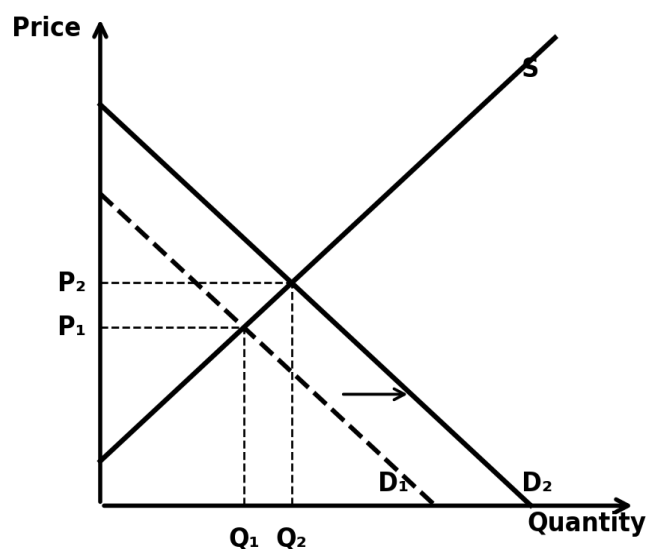
(b) Higher Price of Substitutes → Demand Shifts Right

Figure 1(b): Higher substitute price shifts D rightward — equilibrium price rises ($P_1 \rightarrow P_2$), equilibrium quantity rises ($Q_1 \rightarrow Q_2$).

Full Model Answer

(a) Increase in the Number of Suppliers: An increase in the number of suppliers means more firms are producing the good at every price level. This causes the **supply curve to shift rightward** from S_1 to S_2 , as shown in Figure 1(a). At the original equilibrium price P_1 , there is now excess supply — quantity supplied exceeds quantity demanded. The resulting downward pressure on price moves the market to a new equilibrium where price falls from P_1 to P_2 and quantity rises from Q_1 to Q_2 . More competition in the market drives prices down and increases total output available to consumers.

(b) Increase in the Price of Substitutes: When the price of a substitute good rises, that good becomes relatively more expensive, causing consumers to switch to the original good. This increases demand at every price level, shifting the **demand curve rightward** from D_1 to D_2 , as shown in Figure 1(b). At the original price P_1 , there is now excess demand. Price is bid up from P_1 to P_2 and equilibrium quantity rises from Q_1 to Q_2 . For example, if the price of coffee rises significantly, consumers switch to tea — shifting the demand for tea rightward, raising both its price and quantity sold.

Mark Allocation

Mark Band	Criteria
9–10%	Both parts answered with correct direction of shift clearly stated for each. Both diagrams drawn correctly — S shifts right for (a), D shifts right for (b) — with P and Q effects labelled accurately. Real-world example used for at least one part.
6–8%	Both parts attempted, diagrams mostly correct. Direction of shift correct but price/quantity effects may be incompletely explained.
3–5%	One part answered well. Second part has errors in diagram or direction of shift.
0–2%	No meaningful attempt or fundamental confusion between supply and demand shifts.

A1 (ii)
10 percent

Explain, with the aid of a diagram, the effect on the labour market of a government-imposed minimum wage which is above the market equilibrium wage.

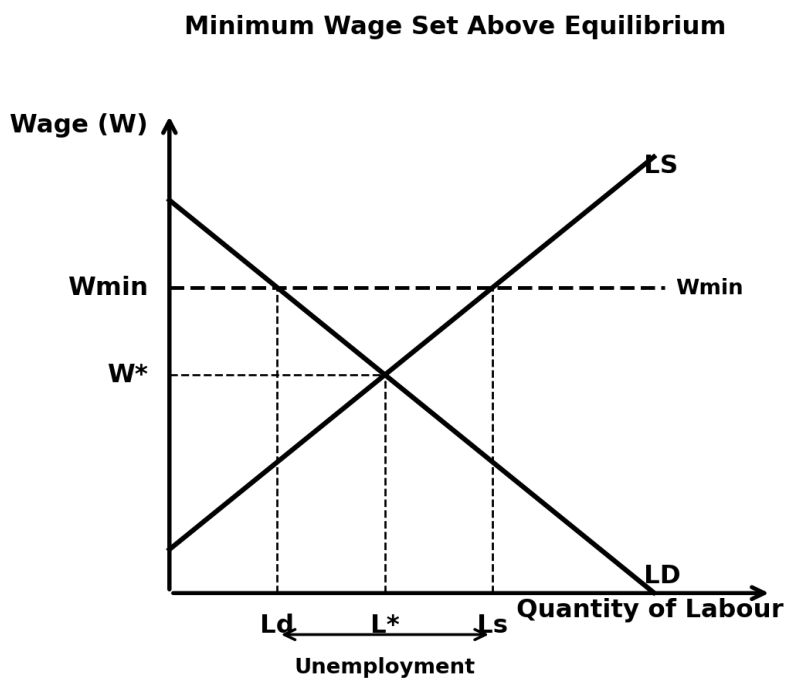


Figure 2: A minimum wage (W_{min}) set above the equilibrium wage (W^*) creates unemployment equal to $L_s - L_d$.

Full Model Answer

Definition: A **minimum wage** is a legally imposed lower bound on the wage that employers may pay workers. It is a type of price floor. When set **above the market equilibrium wage W^*** , it is binding — it actively prevents the labour market from clearing at the competitive equilibrium.

Effect on Labour Demand (Ld): At the higher minimum wage W_{min} , labour becomes more expensive for firms. Employers respond by **demanding less labour** — reducing hours, automating tasks, or cutting staff. Quantity of labour demanded falls from L^* to L_d along the labour demand curve.

Effect on Labour Supply (Ls): The higher wage attracts more workers into the labour market — more people are willing and able to work at W_{min} than at W^* . Quantity of labour supplied rises from L^* to L_s along the labour supply curve.

Unemployment: The gap between L_s and L_d represents **unemployment** — there are more workers seeking jobs at W_{min} than there are jobs available. Unlike a free market, where price adjusts to clear the market, the minimum wage legally prevents wages from falling back toward equilibrium. The surplus of labour ($L_s - L_d$) persists.

Critical Evaluation: While a minimum wage reduces poverty for those who retain employment and raises living standards for low-paid workers, it creates unemployment and may reduce total employment. The size of the unemployment effect depends on the **wage elasticity of labour demand** — if demand is inelastic (e.g. in essential services), unemployment effects are limited. Ireland's National Minimum Wage (€13.50 per hour as of 2024) has been accompanied by broadly stable employment, suggesting elasticity of labour demand is relatively low in the Irish economy (ESRI, 2024).

Mark Allocation

Mark Band	Criteria
9–10%	Correct diagram: LS upward, LD downward, W^* and L^* at equilibrium, W_{min} above W^* , L_d and L_s clearly marked, unemployment bracket shown. Written explanation covers: labour demand falls, labour supply rises, persistent unemployment results. Critical evaluation or real-world example present.
6–8%	Diagram mostly correct. Explanation covers key effects but may lack critical evaluation or real-world context.
3–5%	Basic diagram with some labelling errors. Explanation partially correct — identifies unemployment but does not explain mechanism fully.
0–2%	No meaningful attempt or fundamental confusion with price ceiling.

A2 (i)

10 percent

Describe, using an example, how loss aversion can affect consumers' decisions.

Full Model Answer

Definition: Loss aversion is the well-documented psychological tendency for individuals to feel the pain of a loss more powerfully than the pleasure of an equivalent gain. Formalised by Kahneman and Tversky in Prospect Theory (1979, *Econometrica*), research demonstrates that losses are felt approximately **twice as intensely as equivalent gains** — losing €100 causes roughly twice the psychological pain as gaining €100 causes pleasure.

Consumer Example 1 — Insurance: Loss aversion explains why consumers systematically over-insure — they pay premiums exceeding the actuarial value of the risk because the prospect of a significant loss (house fire, theft, accident) feels disproportionately painful. The disutility of the potential loss looms larger psychologically than the actual probability-weighted cost justifies.

Consumer Example 2 — Marketing and Framing: Firms exploit loss aversion by framing benefits as avoiding a loss rather than making a gain. 'Switch and **SAVE €300**' consistently outperforms 'Switch and **EARN €300 more**', despite being economically identical. Limited-time sale urgency ('You will MISS OUT if you don't act today') leverages loss aversion directly (Cialdini, 1984, *Influence*).

Consumer Example 3 — Endowment Effect: Once a consumer owns something, they value it more highly than before they owned it — making them reluctant to sell or exchange it even at a fair price. This is driven by loss aversion: giving up the owned item feels like a loss, which is psychologically more painful than the equivalent gain from the sale price. Thaler (1980) documented this as the **endowment effect**, a direct consequence of loss aversion.

Mark Allocation

Mark Band	Criteria
9–10%	Accurate definition referenced to Kahneman & Tversky and Prospect Theory. Clear contrast with rational choice assumption of symmetric weighting. Two or more compelling examples with mechanism explained. Policy or commercial application discussed. Citation present.
6–8%	Correct definition and one strong example with mechanism. Limited critical depth.
3–5%	Definition present but Prospect Theory not referenced. One example but mechanism not fully linked to loss aversion.
0–2%	No meaningful attempt or loss aversion confused with other concepts.

A2 (ii)

10 percent

Explain, using an example, how framing can affect consumers' decisions.

Full Model Answer

Definition: Framing refers to the way in which information is presented — the context, wording, or format in which a choice is offered — and how this presentation influences consumer decisions independently of the objective content of the choice. Framing is a core concept in behavioural economics, demonstrating that consumers do not evaluate options in absolute terms but are strongly influenced by how options are described (Tversky & Kahneman, 1981, *Science*).

The Rational Economic Benchmark: Standard economic theory assumes that rational agents evaluate choices based solely on their objective outcomes — the framing of the choice should be irrelevant. If two options have identical expected values, a rational agent should be indifferent between them regardless of how they are presented.

Example 1 — Medical Treatment (Kahneman & Tversky's Classic Experiment): Participants told a medical treatment had a '**90% survival rate**' were significantly more likely to choose it than those told it had a '**10% mortality rate**' — even though both statements are mathematically identical. The gain frame (survival) was far more persuasive than the loss frame (mortality). This demonstrates that consumer decisions are shaped not by the objective data but by how it is presented.

Example 2 — Supermarket Pricing: A yoghurt labelled '**95% fat free**' sells significantly better than the identical product labelled '**contains 5% fat**'. The positive frame triggers more favourable consumer evaluation despite communicating the same nutritional information.

Example 3 — Default Options and Nudging: Framing through default settings is exploited in pension enrolment. When pension membership is framed as the default (opt-out rather than opt-in), participation rates increase dramatically — from around 50% to over 90% in several jurisdictions (Thaler & Sunstein, 2008, *Nudge*). The framing of membership as the standard behaviour powerfully influences consumer decisions.

Mark Allocation

Mark Band	Criteria
9–10%	Clear definition of framing distinct from other biases. Explicit contrast with rational choice theory. Two or more specific examples with mechanism clearly explained. Academic citation (Tversky & Kahneman or Thaler & Sunstein). Critical insight into implications for policy or marketing.
6–8%	Correct definition and one good example with mechanism. Limited critical depth or missing citation.
3–5%	Framing defined but mechanism not applied to example clearly. Example may be vague.
0–2%	No meaningful attempt or framing confused with loss aversion or other concepts.

A3 (i)
10 percent

Using the data in Table 1, calculate the average cost and marginal cost for each unit of output.

Unit	Total Cost (€)	Average Cost (€) Marginal Cost (€)
1	€11	€11.00 €11
2	€20	€10.00 €9
3	€28	€9.33 €8
4	€32	€8.00 €4
5	€35	€7.00 €3
6	€40	€6.67 €5
7	€48	€6.86 €8

Full Model Answer

Formulas:

$$\text{Average Cost (AC)} = \text{Total Cost (TC)} \div \text{Quantity (Q)}$$

$$\text{Marginal Cost (MC)} = \text{Change in Total Cost } (\Delta\text{TC}) \div \text{Change in Quantity } (\Delta\text{Q})$$

Calculations — step by step:

- Unit 1: $\text{AC} = 11 \div 1 = \text{€}11.00$ | $\text{MC} = (11-0) \div 1 = \text{€}11.00$
- Unit 2: $\text{AC} = 20 \div 2 = \text{€}10.00$ | $\text{MC} = (20-11) \div 1 = \text{€}9.00$
- Unit 3: $\text{AC} = 28 \div 3 = \text{€}9.33$ | $\text{MC} = (28-20) \div 1 = \text{€}8.00$
- Unit 4: $\text{AC} = 32 \div 4 = \text{€}8.00$ | $\text{MC} = (32-28) \div 1 = \text{€}4.00$
- Unit 5: $\text{AC} = 35 \div 5 = \text{€}7.00$ | $\text{MC} = (35-32) \div 1 = \text{€}3.00$
- Unit 6: $\text{AC} = 40 \div 6 = \text{€}6.67$ | $\text{MC} = (40-35) \div 1 = \text{€}5.00$
- Unit 7: $\text{AC} = 48 \div 7 = \text{€}6.86$ | $\text{MC} = (48-40) \div 1 = \text{€}8.00$

Key Observation: Marginal cost falls from unit 1 to unit 5 (from €11 to €3) as output increases — reflecting economies from specialisation. It then rises from unit 6 onwards (€5, €8), reflecting diminishing marginal returns as the fixed factor (e.g. factory space or management) becomes increasingly congested. The minimum MC is at unit 5 (€3.00) and minimum AC is at unit 6 (€6.67). Notice MC passes through minimum AC between units 6 and 7 — consistent with the standard relationship where MC intersects AC at its minimum.

Mark Allocation

Mark Band	Criteria
9–10%	Both formulas stated correctly. All 7 AC and MC values calculated accurately and shown with working. Key relationship between MC and AC noted (MC crosses AC at minimum AC).
6–8%	Formulas correct and most values accurate. Minor arithmetic error in 1–2 rows. Working shown.
3–5%	Formulas partially correct or values contain significant errors. Some working present.
0–2%	No meaningful attempt or both formulas incorrect.

A3 (ii)

10 percent

Explain what is meant by economies and diseconomies of scale. Use an example to explain your answer.**Full Model Answer**

Economies of Scale — Definition: Economies of scale occur when a firm's **long-run average cost (LRAC) falls as output increases**. As production expands, the firm becomes more efficient — it can spread fixed costs over more units, access volume discounts, specialise labour, and invest in more productive machinery. The result is that each unit costs less to produce as scale increases.

Sources of Economies of Scale:

- Technical economies: Larger machinery operates at lower cost per unit than multiple smaller machines (e.g. a single large freight ship vs ten small ones).
- Purchasing economies: Bulk buying raw materials secures lower per-unit prices from suppliers.
- Managerial economies: Specialist managers can be employed as scale justifies their cost.
- Financial economies: Larger firms access cheaper borrowing — lower interest rates due to lower default risk.

Real-World Example: Amazon benefits from massive economies of scale in logistics and warehousing. The fixed costs of distribution centres, automated picking systems, and delivery networks are spread across billions of orders annually, making each individual unit cheaper to process and deliver. This allows Amazon to offer lower prices than smaller competitors while maintaining profitability.

Diseconomies of Scale — Definition: Diseconomies of scale occur when a firm's **long-run average cost rises as output increases beyond a certain point**. As firms grow very large, inefficiencies emerge — management becomes harder to coordinate, communication breaks down, bureaucracy increases, and worker motivation can fall. These factors cause average cost to rise despite continued expansion.

Sources of Diseconomies of Scale:

- Managerial diseconomies: Decision-making becomes slow and bureaucratic as layers of management multiply.
- Labour relations: Large organisations may suffer from poor industrial relations, absenteeism, and lower worker motivation.
- Communication failures: Information flows become distorted as the organisation grows.

LRAC Curve: On the Long-Run Average Cost curve, economies of scale are shown by the downward-sloping section, while diseconomies of scale are shown by the upward-sloping section. The lowest point of the LRAC — where average cost is minimised — is the **minimum efficient scale** (Samuelson & Nordhaus, 2010, *Economics*).

Mark Allocation

Mark Band	Criteria
9–10%	Clear accurate definition of both economies and diseconomies of scale in terms of LRAC behaviour. At least two sources of each identified. Specific real-world example with mechanism applied. Reference to minimum efficient scale or LRAC curve. Citation present.
6–8%	Both concepts correctly defined. One or two sources listed. Example present but may lack depth. No citation.
3–5%	One concept well-explained. Second concept partially explained or confused with short-run diminishing returns.
0–2%	No meaningful attempt or fundamental confusion between short-run and long-run cost concepts.

SECTION B — LONG QUESTIONS (Attempt 1 out of 2 · 60% of total marks)

Attempt ONE question from B1 or B2. Each question is worth 60% in total. Each sub-part (i), (ii), (iii) is worth 20%. Target 3–4 pages in total. Include 2 academic references per page.

B1 (i)

20 percent

Using the data in Table 2, calculate the point price elasticity of demand of each item, *ceteris paribus*.

Formula: $PED = \% \text{ Change in Quantity Demanded} \div \% \text{ Change in Price}$

Item	Price 2022	Qty 2022	Price 2023	Qty 2023	PED	Law of Demand?
Shoes	€0.55	93,000	€0.60	85,000	-0.95	No — price ↑, Qd ↓
Bag	€0.85	72,000	€1.05	50,000	-1.29	Yes — PED > 1 elastic
Socks	€0.87	110,000	€0.75	115,000	+0.30	No — Giffen-like (↑P, ↑Qd)
Jewellery	€1.35	95,000	€1.25	85,000	+1.35	No — P ↓, Qd ↓ (Veblen-type)

Full Model Answer

Worked Calculations:

Shoes: $\% \Delta Q_d = (85-93) \div 93 \times 100 = -8.60\%$ | $\% \Delta P = (0.60-0.55) \div 0.55 \times 100 = +9.09\%$ | $PED = -8.60 \div 9.09 = -0.95$

Bag: $\% \Delta Q_d = (50-72) \div 72 \times 100 = -30.56\%$ | $\% \Delta P = (1.05-0.85) \div 0.85 \times 100 = +23.53\%$ | $PED = -30.56 \div 23.53 = -1.30$

Socks: $\% \Delta Q_d = (115-110) \div 110 \times 100 = +4.55\%$ | $\% \Delta P = (0.75-0.87) \div 0.87 \times 100 = -13.79\%$ | $PED = +4.55 \div -13.79 = -0.33$

Jewellery: $\% \Delta Q_d = (85-95) \div 95 \times 100 = -10.53\%$ | $\% \Delta P = (1.25-1.35) \div 1.35 \times 100 = -7.41\%$ | $PED = -10.53 \div -7.41 = +1.42$

Note on Socks and Jewellery: Socks show price falling but quantity rising — a negative PED suggesting normal demand behaviour, inelastic. Jewellery shows both price and quantity falling simultaneously — a positive PED value, suggesting the good may behave as a Veblen good (demand falls when price falls, as lower price reduces prestige appeal). Students should note these anomalies and explain them.

Mark Allocation

Mark Band	Criteria
17–20%	Formula stated. All four calculations shown step by step with $\% \Delta Q_d$ and $\% \Delta P$ calculated separately before division. All four PED values correct or very close (within rounding). Anomalous results for Socks and Jewellery noted and discussed.
12–16%	Formula correct. Three or four calculations attempted with correct method. Minor arithmetic errors. Working shown clearly.
7–11%	Formula correct. Two calculations completed correctly. Method present but significant arithmetic errors in others.
3–6%	Formula partially correct or working absent. At least one value approximately correct.
0–2%	No meaningful attempt or completely wrong formula.

B1 (ii)

20 percent

Identify the good(s) listed in Table 2 which follow the Law of Demand and reflect a relatively elastic price elasticity of demand. Explain your answer.

Full Model Answer

Step 1 — Which goods follow the Law of Demand?

The Law of Demand states that, ceteris paribus, as price rises, quantity demanded falls — a negative price-quantity relationship (negative PED). Checking each item:

- **Shoes:** Price rose (€0.55→€0.60), Quantity fell (93→85). ✓ Follows Law of Demand. PED = -0.95.
- **Bag:** Price rose (€0.85→€1.05), Quantity fell (72→50). ✓ Follows Law of Demand. PED = -1.30.
- **Socks:** Price fell (€0.87→€0.75), Quantity rose (110→115). ✓ Follows Law of Demand (price down, quantity up). PED = -0.33.
- **Jewellery:** Price fell (€1.35→€1.25), Quantity fell (95→85). ✗ Does NOT follow Law of Demand — price and quantity moved in the same direction (both fell). PED = +1.42, suggesting Veblen good behaviour.

Step 2 — Which of the Law-of-Demand goods are relatively elastic ($|PED| > 1$)?

- Shoes: $|PED| = 0.95 < 1 \rightarrow$ relatively **inelastic**. Not elastic.
- Bag: $|PED| = 1.30 > 1 \rightarrow$ relatively **elastic**. ✓ This is the answer.
- Socks: $|PED| = 0.33 < 1 \rightarrow$ relatively **inelastic**. Not elastic.

Answer: The **Bag** is the only good that both follows the Law of Demand (PED negative, price and quantity move in opposite directions) **and** has a relatively elastic demand ($|PED| = 1.30 > 1$).

Interpretation of Bag's Elasticity: A PED of -1.30 means that a 1% increase in the price of the bag leads to a 1.30% fall in quantity demanded. The demand is elastic because bags are likely to have reasonable substitutes (other brands, different styles, second-hand options), are not necessities, and represent a significant share of consumer spending — all factors that increase price sensitivity.

Mark Allocation

Mark Band	Criteria
17–20%	Correct identification of Bag as the only good satisfying both conditions, with clear reasoning referencing both the negative PED criterion (Law of Demand) and the elastic criterion ($ PED > 1$). Each good addressed individually. Jewellery identified as anomalous. Explanation of why Bag is elastic (substitutes, non-necessity etc.).

Mark Band	Criteria
12–16%	Bag correctly identified. Most goods correctly assessed against Law of Demand and elasticity criteria. Jewellery anomaly may not be fully explained.
7–11%	Bag identified but reasoning incomplete. Some goods incorrectly classified.
3–6%	Partial attempt — correct identification of some but not all criteria.
0–2%	No meaningful attempt or Law of Demand misunderstood.

B1 (iii)
20 percent**Discuss factors that can determine the magnitude of the price elasticity of demand of goods and services.**

Full Model Answer

Introduction: The magnitude of the price elasticity of demand (PED) — how strongly consumers respond to a price change — is not fixed. It varies across goods and markets depending on several key determinants. Understanding these factors is essential for firms setting prices and governments designing taxation policy.

1. Availability of Close Substitutes

This is the single most important determinant of PED. The more close substitutes a good has, the more elastic demand will be, because consumers can easily switch to an alternative when the price rises. A branded good like Coca-Cola has more elastic demand than petrol because Pepsi is an easy substitute. Goods with few or no substitutes — prescription medication, insulin, utilities — have highly inelastic demand because consumers cannot switch away when prices rise (Lipsey & Chrystal, 2015, *Economics*).

2. Necessity vs Luxury

Necessities tend to have inelastic demand — consumers must purchase them regardless of price increases. Food staples, basic clothing, medication, and utilities all show low PED. Luxuries, by contrast, tend to have more elastic demand — consumers can postpone, reduce, or forgo luxury purchases when prices rise. A price rise in foreign holidays will reduce demand significantly, whereas a price rise in bread will not. The distinction explains why governments often impose higher taxes on luxuries (alcohol, tobacco, fuel) than on food staples.

3. Proportion of Income Spent on the Good

Goods that represent a large share of consumer income tend to have more elastic demand. A 10% rise in the price of a car — which costs €30,000 — causes consumers to think very carefully before purchasing. A 10% rise in the price of a pen — costing €1 — is barely noticed and has almost no effect on demand. This explains why demand for expensive durable goods (cars, appliances, housing) is typically more elastic than demand for cheap everyday items (salt, matches, newspapers).

4. Time Horizon

Demand tends to be more elastic in the long run than the short run. In the short run, consumers may be locked into habits, contracts, or equipment that prevents them from switching. Over time, they can find substitutes, change behaviour, or invest in alternatives. For example, in the short run a petrol price rise has limited effect on demand because most drivers are committed to petrol-powered cars. In the long run, however, sustained high petrol prices drive consumers to switch to electric vehicles, public transport, or fuel-efficient cars — making demand significantly more elastic (Nicholson & Snyder, 2012, *Microeconomic Theory*).

5. Breadth of Market Definition

The more narrowly a market is defined, the more elastic demand tends to be. Demand for 'food' as a category is highly inelastic — no substitutes exist. Demand for 'Tayto crisps' is highly elastic

because many substitute crisps exist. Demand for 'Mars bars specifically' is more elastic than demand for 'chocolate bars generally'. Market definition therefore crucially affects measured PED.

6. Habit and Addiction

Addictive goods — tobacco, alcohol, certain medications — tend to have inelastic demand because the habituated consumer finds it difficult to reduce consumption even when prices rise significantly. Habitual purchases such as morning coffee or regular newspaper subscriptions also show lower price sensitivity than considered one-off purchases.

Mark Allocation

Mark Band	Criteria
17–20%	Five or more factors covered, each with: definition, explanation of mechanism (why this factor makes demand more or less elastic), and a specific real-world example. Critical analysis present — e.g. comparing factors or noting interactions. Multiple academic citations.
12–16%	Three or four factors well-explained with examples. Some critical analysis. At least one citation.
7–11%	Two or three factors listed and partially explained. Examples may be vague.
3–6%	One or two factors mentioned without sufficient explanation or examples.
0–2%	No meaningful attempt.

B2 (i)
20 percent

Explain, using an illustration, how price and output are determined for a perfectly competitive firm.

Perfect Competition — Short Run (Supernormal Profit)

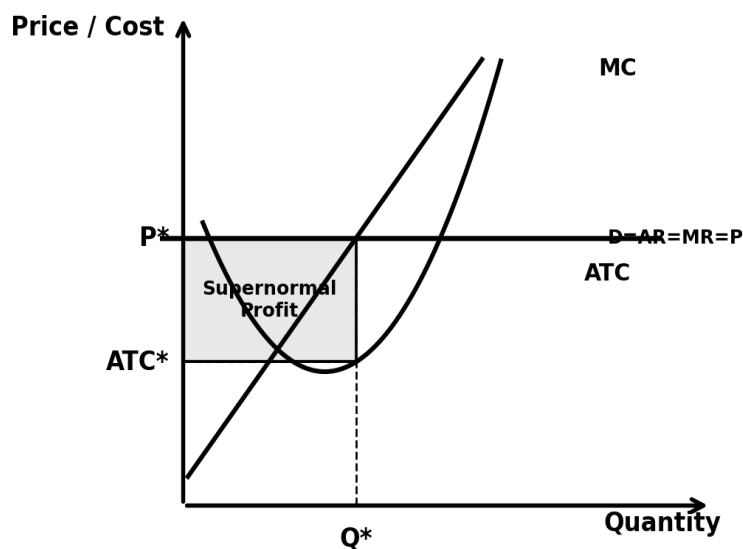


Figure 3: Perfectly competitive firm — short run. The firm is a price taker: price P^* is set by the market. Profit is maximised at Q^* where $MC = MR = P^*$. If $P^* > ATC^*$, the firm earns supernormal profit (shaded).

Full Model Answer

Market Structure: Perfect competition is a market structure characterised by: **many buyers and sellers; homogeneous (identical) products; perfect information;** and completely **free entry and exit**. No single buyer or seller can influence the market price. Real-world approximations include agricultural commodity markets (wheat, milk) and foreign exchange markets.

The Price Taker: A perfectly competitive firm is a **price taker** — it must accept the price determined by the market's aggregate supply and demand. Since each firm's output is an infinitesimally small fraction of total market supply, no individual firm can influence price by changing its output. The firm faces a perfectly elastic (horizontal) demand curve at the market price P^* .

Price = MR = AR: For a price-taking firm, the demand curve ($D = AR$) is horizontal at P^* . Since the firm sells every unit at the same price P^* , its Marginal Revenue (MR) also equals P^* — each additional unit sold adds exactly P^* to total revenue. Therefore: **$P^* = MR = AR$** . This is the crucial distinction from monopoly — in perfect competition, $MR = P$.

Profit Maximisation — $MC = MR$: The firm maximises profit by producing the output level where **Marginal Cost (MC) = Marginal Revenue (MR)**. Since $MR = P^*$ in perfect competition, this becomes $MC = P^*$. At output Q^* (as shown in Figure 3), the firm sets $MC = MR = P^*$. Producing less than Q^* means $MR > MC$ — more output would add more to revenue than to cost, so more should be produced. Producing more than Q^* means $MC > MR$ — each additional unit costs more than it adds in revenue, so output should be reduced.

Short-Run Profit Outcomes: The firm's profit position depends on the relationship between P^* and Average Total Cost (ATC^*) at Q^* :

- **If $P^* > ATC^*$:** Supernormal profit (as shown in Figure 3 — the shaded rectangle represents profit per unit $\times$ quantity).
- **If $P^* = ATC^*$:** Normal profit — the firm covers all costs including a normal return on capital but earns nothing above this.
- **If $P^* < ATC^*$:** Loss — the firm loses money. It should continue operating in the short run as long as $P^* > AVC$ (price covers variable costs), as shutting down would incur greater fixed cost losses.

Long-Run Equilibrium: In the long run, free entry and exit ensures that supernormal profits are competed away (new entrants drive price down) and losses lead to exit (fewer firms drives price up) until **$P^* = MC = \text{minimum } ATC$** — where the firm earns normal profit only, the market is allocatively efficient, and the industry is in equilibrium.

Mark Allocation

Mark Band	Criteria
17–20%	Diagram correctly drawn: horizontal $D=AR=MR=P$ line, upward MC, U-shaped ATC, Q^* where $MC=MR$, profit rectangle or loss shown. Written explanation covers: price-taking nature, $MR=P=AR$, $MC=MR$ profit maximisation rule, all three profit outcomes, and long-run adjustment. Academic reference or real-world example present.

Mark Band	Criteria
12–16%	Diagram present with most labels. Price-taking and $MC=MR$ rule explained. Long-run adjustment or profit outcomes may lack depth.
7–11%	Basic diagram. Core price-taking concept explained but $MC=MR$ rule not fully applied or profit analysis incomplete.
3–6%	Partial diagram or explanation. Significant errors or omissions.
0–2%	No meaningful attempt.

B2 (ii)
20 percent

Explain, using diagrams, how profits earned by perfectly competitive firms may differ in the short and long run.

Short Run — Supernormal Profit:

Perfect Competition — Short Run (Supernormal Profit)

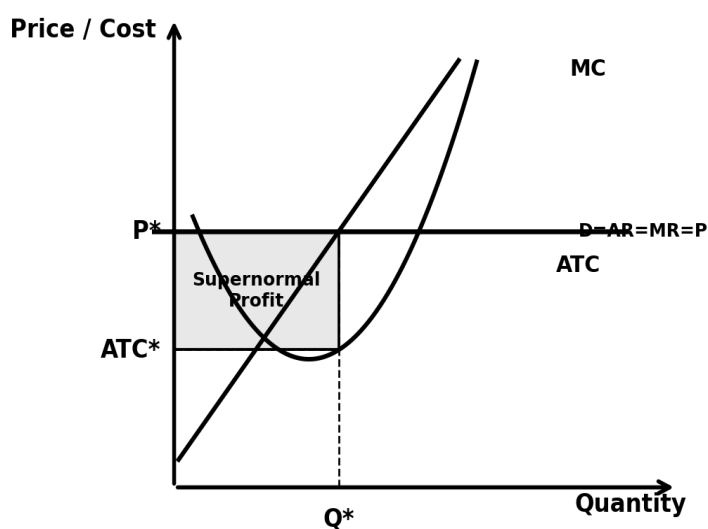


Figure 4: Short-run supernormal profit. $P^* > ATC^*$ at Q^* ($MC=MR$). Shaded rectangle = profit.

Short Run — Loss:

Perfect Competition — Short Run (Loss)

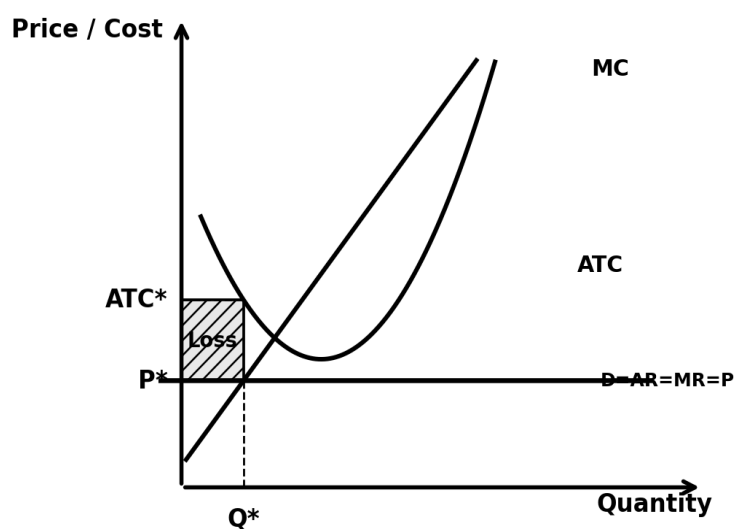


Figure 5: Short-run loss. $P^* < ATC^*$ at Q^* ($MC=MR$). Hatched rectangle = loss.

Long Run — Normal Profit:

Perfect Competition — Long Run (Normal Profit)

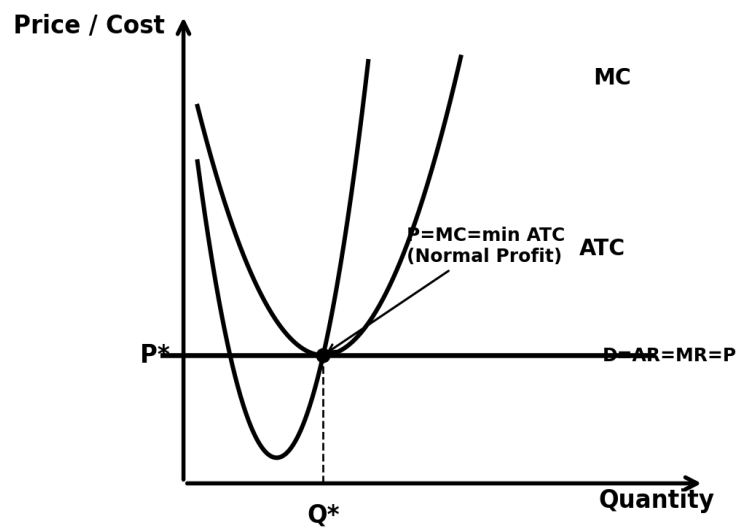


Figure 6: Long-run equilibrium. $P^* = MC = \min ATC$. Normal profit only. Entry/exit has eliminated supernormal profit or loss.

Full Model Answer

Why Short-Run Profits Can Vary: In the short run, the number of firms in the market is fixed — barriers to instant entry or exit mean that market price may temporarily lie above or below Average Total Cost. This creates three possible short-run outcomes, all shown in Figures 4 and 5 above.

Short-Run Supernormal Profit (Figure 4): When market demand is high relative to current supply, the market price P^* may exceed ATC^* . In this case the firm earns **supernormal profit** — a return above the normal competitive return. The profit per unit is $(P^* - ATC^*)$ and total profit is this margin times Q^* . This might arise from a sudden increase in market demand (e.g. exceptional weather boosting demand for ice cream) or from a temporary reduction in industry supply.

Short-Run Loss (Figure 5): When market price falls below ATC^* , the firm makes a **loss**. However, if P^* still exceeds Average Variable Cost (AVC), the firm should continue producing in the short run — it is covering its variable costs and contributing to fixed costs. Shutting down would mean losing all fixed costs with zero revenue. Only if $P^* < AVC$ should the firm shut down immediately.

Long-Run Adjustment — Entry and Exit: The defining feature of perfect competition is **free entry and exit**. This ensures that short-run profits and losses are temporary:

- Supernormal profit attracts new entrants → market supply increases → market price falls → profit is competed away.
- Losses cause existing firms to exit → market supply falls → market price rises → losses are eliminated.

Long-Run Equilibrium (Figure 6): Entry and exit continue until **$P^* = MC = \text{minimum ATC}$** . At this point the firm earns only **normal profit** — just enough to keep it in the industry but no more. The market is both allocatively efficient ($P = MC$) and productively efficient (production at minimum ATC). This long-run equilibrium is the hallmark of perfect competition (Marshall, 1890; Varian, 2014, *Intermediate Microeconomics*).

Mark Allocation

Mark Band	Criteria
17–20%	All three diagrams present and correctly drawn — SR supernormal profit ($P > ATC$, shaded), SR loss ($P < ATC$, hatched), LR normal profit ($P = MC = \text{min} ATC$, tangency). Written explanation covers: why SR profits vary, entry/exit mechanism, and precise LR equilibrium condition. Reference to shutdown condition ($P < AVC$). Academic citation present.
12–16%	Two diagrams correct. SR outcomes explained. Entry/exit mechanism described. LR equilibrium condition stated but may lack depth.
7–11%	One diagram correct. SR and LR distinguished but entry/exit mechanism incomplete.
3–6%	Basic diagram only. SR and LR not clearly distinguished.
0–2%	No meaningful attempt.

B2 (iii)
20 percent

Explain how the role of product differentiation and advertising varies in different market types.

Full Model Answer

Introduction: Product differentiation is the process of making a good or service distinct from competitors' offerings in the eyes of consumers. Advertising is one means of achieving differentiation or signalling it. The extent to which firms engage in differentiation and advertising varies fundamentally across market structures, reflecting differences in market power, the nature of products, and the intensity of competition.

1. Perfect Competition — No Differentiation, No Advertising

In perfect competition, products are by definition **perfectly homogeneous** — identical in every respect across all suppliers. Since consumers can see through any claimed difference and switch instantly to any alternative supplier at the same price, there is **no incentive for product differentiation or advertising**. Individual firms are price takers — they face a perfectly elastic demand curve and any expenditure on advertising would simply raise costs without generating additional demand or pricing power. Agricultural commodity markets (wheat, milk, crude oil) approximate this — a farmer has no reason to advertise because their product is identical to every other farmer's output and price is set by the market.

2. Monopolistic Competition — Active Differentiation, Significant Advertising

Monopolistic competition is defined by differentiation. Each firm produces a product that is a close but **imperfect substitute** for rivals' products. Differentiation — through quality, branding, design, location, or customer service — is the primary source of each firm's pricing power and downward-sloping demand curve. Without differentiation, the market would collapse to perfect competition and the firm would lose all pricing power.

Advertising in monopolistic competition serves to: (a) signal the existence and attributes of the differentiated product to consumers; (b) build brand loyalty, reducing demand elasticity; and (c) attract consumers away from rivals' close substitutes. The restaurant, hairdressing, and clothing retail industries are archetypal examples — firms invest heavily in branding, social media marketing, and design to differentiate their offering (Chamberlin, 1933, *The Theory of Monopolistic Competition*).

3. Oligopoly — Strategic Differentiation, Intensive Advertising

In oligopoly, a small number of large firms compete intensively. Differentiation and advertising are particularly prominent because firms are strategically interdependent — each firm's advertising and product decisions directly affect rivals, who respond in kind. This leads to non-price competition as the primary battleground, since price competition can trigger destructive price wars.

The Coca-Cola vs Pepsi rivalry is the classic example — both firms spend billions annually on advertising despite selling nearly identical products. The differentiation is largely achieved through branding, packaging, sponsorship, and cultural association rather than fundamental product differences. In the smartphone oligopoly (Apple, Samsung, Google), differentiation is both real (hardware and software differences) and heavily advertised. Large advertising expenditures in oligopolies act as a **barrier to entry** — new entrants cannot easily match the brand recognition built by incumbents over decades.

4. Monopoly — Minimal Advertising, No Competition-Driven Differentiation

A pure monopolist faces no competitors — by definition there is only one supplier. There is therefore no competitive pressure to differentiate from rivals (there are none) or to advertise to attract consumers away from competitors. However, monopolists may still advertise to: (a) **expand total market demand** — encouraging non-consumers to enter the market; (b) **deter potential entrants** — signalling market strength; or (c) improve public relations. Natural monopolies (utilities, rail networks) rarely advertise their core service for competitive purposes, though they may advertise ancillary products.

Critical Evaluation: The social welfare implications of advertising and differentiation are contested. From an efficiency perspective, advertising expenditure in monopolistic competition and oligopoly may represent a social waste — resources spent shifting demand between rivals without expanding the total economic pie (Dixit & Stiglitz, 1977). However, advertising provides consumers with information, funds media and content, and — in the case of genuine product differentiation — creates real consumer value through variety (Lancaster, 1979). The net welfare impact depends on whether advertising is primarily informative or merely persuasive.

Mark Allocation

Mark Band	Criteria
17–20%	All four market structures covered with: role of differentiation explained, role of advertising explained, specific real-world example, and why this level of differentiation/advertising arises from the structure's characteristics. Critical evaluation of social welfare implications. Multiple citations. Coherent comparative analysis.
12–16%	Three market structures well-covered. Differentiation and advertising roles explained with examples. Limited critical evaluation.
7–11%	Two market structures compared. Differentiation and advertising discussed but analysis may be superficial.
3–6%	One or two market types mentioned without systematic analysis.
0–2%	No meaningful attempt.